Muhammad Tehmasib Ali Tashfeen

PhD Candidate · Electrical Engineering & Computer Science

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Education

PhD **Wichita State University**, Electrical Engineering and Computer Science — Wichita, Kansas, USA, Aug 2023 – May 2027

- Dissertation (in progress): "Automated Detection of Android Lifecycle Anti-Patterns: From Static Analysis to Fine-Tuned Large Language Models"
- Advisor: Dr. Zhiyong Shan, Software Analysis and Intelligence Laboratory (NSF-supported)
- Research focus: AI-assisted software engineering, static program analysis, Android application reliability and security, domain adaptation of code language models, empirical software engineering

BS **National University of Computer and Emerging Sciences (FAST-NUCES)**, Electrical Engineering — Peshawar, Pakistan, Aug 2015 – July 2019

- CGPA: 3.41 / 4.00
- Senior Design Project: "IoT-Based Smart Metering System"
- Dean's List — Fall 2016, Fall 2017, Fall 2018

Research Interests

- AI-assisted software engineering; static program analysis; Android application reliability and security; domain adaptation of code language models; empirical software engineering and reproducible evaluation.

Experience

Wichita State University — Software Analysis & Intelligence Laboratory (NSF-supported), Graduate Research Assistant — Wichita, Kansas, USA, Aug 2023 – present, 3 years 4 months

- Advisor: Dr. Zhiyong Shan
- Designed and implemented LeakScope, a Soot-based framework that translates Android lifecycle obligations into rules for detecting memory-retention and resource-management patterns in compiled applications
- Evaluated the framework on 627 open-source Android apps, flagging 485 apps in the reported study; compared app-level detection coverage with Android Lint, SpotBugs, and Infer
- Constructed 3,152 prompt-response examples and developed LoRA fine-tuning pipelines for CodeLlama, Llama-3, and StarCoder2, evaluated against GPT-4o; produced research software and benchmark data
- Leveraged the NSF ACCESS / Expanse HPC cluster (Slurm, GPU jobs) for large-scale model training and evaluation
- Lead ongoing work on context-aware fine-tuning of open-source language models for Android anti-pattern detection, with a manuscript in preparation
- Mentors junior graduate and undergraduate researchers; contributes to paper writing, experimental design, and reproducibility

Wichita State University — Software Analysis & Intelligence Laboratory, Research Mentor — Wichita, Kansas, USA
Jan 2026 – present
11 months

- Mentored one undergraduate student researcher and one junior PhD student on static program analysis, empirical software engineering methods, and evaluation workflows
- Supervised ground-truth defect labeling, code inspection criteria, and inter-rater quality control procedures for Android lifecycle memory-retention benchmarks
- Mentorship directly supported the student's research contributions, resulting in their co-authorship on the LeakScope manuscript in preparation for MobiSys 2027

Ernst & Young Ford Rhodes, Senior Consultant, Technology Risk — Islamabad, Pakistan
Mar 2023 – June 2023
4 months

- Led cybersecurity audits for Pakistan Telecommunication Company Ltd. (PTCL) and Ufone, assessing data centers, network security, and SOC operations
- Evaluated security controls and Zero Trust principles; translated findings into prioritized remediation roadmaps
- Prepared executive-facing audit observations connecting infrastructure weaknesses to operational risk and security-control maturity

The Bank of Punjab, Information Systems Auditor — Lahore, Pakistan
Nov 2020 – Mar 2023
2 years 5 months

- Conducted security audits across critical banking systems and evaluated 100+ applications for vulnerabilities and regulatory compliance
- Prepared quarterly Information Security Assurance reports for the Board of Governors, including risk heat maps and remediation timelines
- Assessed access control, change management, incident response readiness, and application security controls under State Bank of Pakistan governance requirements and PCI DSS

Iron Bridge Systems, Executive Engineer — Lahore, Pakistan
Jan 2020 – Nov 2020
11 months

- Deployed and configured enterprise security infrastructure including firewalls, SIEM, WAF, and endpoint protection for client environments
- Validated client-facing security controls for monitoring, prevention, and incident response readiness across multiple engagements

Wichita State University, School of Computing, Graduate Teaching Assistant — Advanced Algorithms/Analysis (CS 721) — Wichita, Kansas, USA
Jan 2026 – May 2026
5 months

- Tuesday / Thursday 14:00 · Section 23280
- Supported instruction in algorithm design, asymptotic complexity, divide-and-conquer, dynamic programming, graph algorithms, and NP-completeness; evaluated coursework and examinations

Wichita State University, School of Computing, Graduate Teaching Assistant — Introduction to Programming (CS 211) — Wichita, Kansas, USA
Jan 2026 – May 2026
5 months

- Tuesday / Thursday 09:30 · Section 23251
- Led two weekly C++ lab sessions covering control flow, functions, data structures, modular design, and object-oriented programming; supported grading and student communication
- Guest instruction (Spring 2026): delivered lectures on cross-platform C++ development, installation, and deployment across Windows, macOS, and Linux

Teaching Areas

- **Foundational computing:** introductory programming (C++, Python, Java), data structures and algorithms, and software engineering.
- **Specialized courses and seminars:** computer security, Android application development and mobile security, static program analysis, and AI for software engineering.

Honors and Awards

Award for Excellence in Research & Innovation Sept 2026
Awarded September 30, 2026.

Research Excellence Award — Excellence in Research and Innovation Sept 2026

Young Researcher Award Oct 2025
Awarded for exceptional dedication and promise in research; presented for outstanding early-career achievements. Proceedings published by Springer, Information Systems Engineering and Management series (vol. 86).

Research Excellence Award Oct 2025
Recognized for high quality, originality, and impactful contribution to Computing, Artificial Intelligence, and Emerging Technologies. Proceedings published by Springer (vol. 86).

Jayne and Glenn Milburn Fellowship Jan 2026
2026–2027 — $400 graduate fellowship awarded based on academic eligibility and financial need.

ISSRE 2026 Travel Funding Sept 2026
Approved allocation of $1,500, subject to reimbursement requirements, to present at the 37th IEEE International Symposium on Software Reliability Engineering (ISSRE 2026).

Student Travel Grant Oct 2024
Awarded travel support to present research on SDN side-channel vulnerabilities at the University of Oklahoma, Norman, Oklahoma.

Professional Service

Peer Reviewer: IEEE Communications Standards Magazine, 2026 — completed review of one manuscript. Public Researcher Profile

Professional Membership: Member, Institute of Electrical and Electronics Engineers (IEEE)

Conference Volunteering: Volunteer, IEEE Region 5 Conference 2025, Wichita State University, March 2025

Student Leadership: Treasurer, Pakistani Student Association, Wichita State University, 2023–2025; Social Media Head, IEEE Student Branch, FAST-NUCES Peshawar Campus

Certificates

Certified Ethical Hacker (CEHv11)

Certified in Cybersecurity (CC)

Talks and Presentations

- **Bridging Program Analysis and Large Language Models for Mobile App Security and Reliability** — Keynote Address, International Conference on Artificial Intelligence, Cyber Security, Multimodal Robotics, and Drones (ACMRD 2026), Uelzen & Berlin, Germany (online), September 5–12, 2026. Slides
- **Toward Reliable Detection of Android Lifecycle Violations with Static Analysis and Domain-Adapted Code Models** — Doctoral Symposium presentation, 37th IEEE International Symposium on Software Reliability Engineering (ISSRE 2026), Limassol, Cyprus, October 2026. Slides
- **LeakScope: A Research Artifact for Android Lifecycle Violations** — Tool pitch presentation, ISSRE 2026, Limassol, Cyprus, October 2026.
- **Toward Reliable Detection of Android Lifecycle Violations with Static Analysis and Domain-Adapted Code Models** — Poster presentation, ISSRE 2026, Limassol, Cyprus, October 2026.
- **Automated Detection of Android Lifecycle Anti-Patterns: From Static Analysis to Fine-Tuned Large Language Models** — PhD Dissertation Proposal Defense, School of Computing, Wichita State University, May 11, 2026. Report
- Three Minute Thesis (3MT) Competition — Participant, Wichita State University Graduate School, April 25, 2026.
- PhD Qualifying Examination — Electrical Engineering and Computer Science, Wichita State University, October 2025. Slides Report
- **Software-Based Side-Channel Attacks in SDNs** — Oral presentation, 17th Central Area Networking and Security Workshop (CANSec 2024), University of Oklahoma, Norman, Oklahoma, October 2024. With Z. Shan. Slides
- **Timing Side-Channel Attacks and Defenses for Login Authentication** — Technical workshop/demo, WuLUG, February 2024. Demonstrated timing side-channel risk in password-checking logic and presented mitigation code for constant-time comparison and secure authentication handling. Slides Demo Script
- **Software-Based Side-Channel Attacks** — Poster presentation, College of Engineering Research Day, Wichita State University, John Bardo Center, November 2023. With Z. Shan. Poster

Skills

Program Analysis

Mobile & Android Security

Machine Learning & LLMs

Systems & Security

Programming

Research Computing

Languages

English
Professional working proficiency

Urdu
Native proficiency